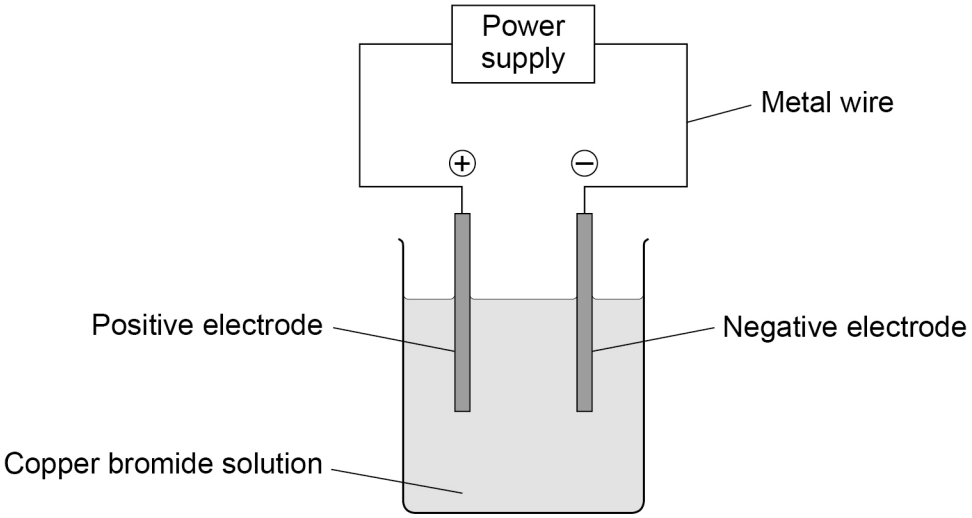
Do not write outside the box

0 4 Copper bromide solution is electrolysed using inert electrodes.

Figure 11 shows the apparatus.

Figure 11



0 4 . 1 Which particles carry the electrical charge through the metal wire?

[1 mark]

Tick (✓) **one** box.

Electrons ☐

Neutrons ☐

Protons ☐

Question 4 continues on the next page

Do not write outside the box

There are four ions in copper bromide solution:

- Cu^{2+}
- Br^-
- H^+
- OH^-

0 4 . 2 Two of these ions are formed when a water molecule breaks down.

The symbol equation when a water molecule breaks down is:

$$H_2O \rightarrow H^+ + OH^-$$

Complete the **word** equation for the breakdown of a water molecule.

[2 marks]

water → _____ ion + _____ ion

0 4 . 3 Copper ions and bromide ions carry the electrical charge through the solution.

The formula of a copper ion is Cu^{2+}

The formula of a bromide ion is Br^-

What is the formula of copper bromide?

[1 mark]

Tick (✓) **one** box.

CuBr ☐

Cu_2Br ☐

$CuBr_2$ ☐

Do not write outside the box

0 4 . 4 Explain why copper ions (Cu^{2+}) move to the negative electrode.

[2 marks]

_____

_____

_____

_____

0 4 . 5 Complete the sentence.

Choose the answer from the box.

[1 mark]

decomposed	**discharged**	**distilled**

At the negative electrode copper metal is produced when the

copper ions are _____ .

0 4 . 6 What happens to the mass of the **negative** electrode during electrolysis?

[1 mark]

Tick (✓) **one** box.

Decreases ☐

No change ☐

Increases ☐

Do not write outside the box

There are four ions in copper bromide solution:

- Cu^{2+}
- Br^{-}
- H^{+}
- OH^{-}

0 4 . 7 What is produced at the **positive** electrode when copper bromide solution is electrolysed?

[1 mark]

Tick (✓) **one** box.

Bromine ☐

Hydrogen ☐

Oxygen ☐

9